

UNIT-5

Point estimation

Basic definitions

- **Population:** The group of individual under study is called Population.
- **Sample:** A finite subset of statistical individuals in a population is called Sample.
- **Sample size:** The number of individuals in a sample is called sample size.
- **Simple Random Sampling:** A random sample is one in which each unit of population has an equal chance (say p) of being included in it and this probability is independent of the previous drawing.
- **Parameter:** The statistical constant of population is called parameter e.g. Mean μ , Variance σ^2 etc.

Basic definitions

- **Statistics:** Statistical measures computed from the sample observations e.g. sample mean $\bar{x}$, sample variance s^2 etc.
- **Estimator:** Any function of the random sample $x_1, x_2, \dots, x_n$ that are being observed, say $T_n(x_1, x_2, \dots, x_n)$ is called a statistic, it is a random variable and it is used to estimate an unknown parameter θ of the distribution, it is called an estimator. A particular value of the estimator say $T_n(x_1, x_2, \dots, x_n)$ is called an estimate of θ .

Characteristics of Estimators

- Unbiasedness
- Consistency
- Efficiency
- Sufficiency

Unbiasedness

- **Unbiased Estimator:** An estimator $T_n = T(x_1, x_2, \dots, x_n)$ is said to be an unbiased estimator of parameter $\gamma(\theta)$, if
$$E(T_n) = \gamma(\theta)$$

Q: In normal distribution $N(\mu, \sigma^2)$, show that sample mean $\bar{x}$ is an unbiased estimator of population mean μ , but sample variance s^2 is biased estimator of population variance σ^2 .

Q: If $x_1, x_2, \dots, x_n$ is a random sample from normal population $N(\mu, 1)$. Show that $t = \frac{1}{n} \sum_{i=1}^n x_i^2$ is an unbiased estimator of $\mu^2 + 1$.

Practice Questions

Q: If T is an unbiased estimator for θ , show that T^2 is a biased estimator for θ^2

Q. If $x_1, x_2, \dots, x_n$ be a random sample of size n from Normal population with mean μ and variance σ^2 then check the biasedness of $t = \frac{1}{n} \sum_{i=1}^n x_i^2$ over μ^2 .

Q. If X is a binomial random variable then discuss the biasedness of $\frac{X}{n}$ and $\frac{X + \frac{\sqrt{n}}{2}}{n + \sqrt{n}}$ over population parameter p .

Practice Questions

Example 15.3. Show that $\frac{[\sum x_i (\sum x_i - 1)]}{n(n-1)}$ is an unbiased estimate of θ , for the sample $x_1, x_2, \dots, x_n$ drawn on X which takes the values 1 or 0 with respective probabilities θ and $(1 - \theta)$.

Example 15.4. Let X be distributed in the Poisson form with parameter θ . Show that the only unbiased estimator of $\exp [-(k+1)\theta]$, $k > 0$, is $T(X) = (-k)^X$ so that

$$T(x) > 0 \text{ if } x \text{ is even}$$

and

$$T(x) < 0 \text{ if } x \text{ is odd.}$$

Consistency

- An estimator $T_n = T(x_1, x_2, \dots, x_n)$ based on random sample of size n , is said to be consistent estimator of $\gamma(\theta)$, if T_n converges to $\gamma(\theta)$ in probability, i.e. $T_n \xrightarrow{p} \gamma(\theta)$ as $n \rightarrow \infty$.
- In other words, T_n is consistent estimator of $\gamma(\theta)$ if for every $\epsilon > 0$, $\eta > 0$, there exists a positive integer $n \geq m(\epsilon, \eta)$ such that

$$\begin{aligned} P\{|T_n - \gamma(\theta)| < \epsilon\} &\rightarrow 1 \text{ as } n \rightarrow \infty \\ \Rightarrow P\{|T_n - \gamma(\theta)| < \epsilon\} &> 1 - \eta \quad \forall n \geq m \end{aligned}$$

Where m is very large value of n .

Invariance property of consistent estimator

- If T_n is a consistent estimator of $\gamma(\theta)$ and $\psi\{\gamma(\theta)\}$ is a continuous function of $\gamma(\theta)$, then $\psi(T_n)$ is a consistent estimator of $\psi\{\gamma(\theta)\}$.

Sufficient conditions for consistency

- Let $\{T_n\}$ be a sequence of estimators, such that $\forall \theta \in \Theta$

(i) $E(T_n) \rightarrow \gamma(\theta)$, as $n \rightarrow \infty$

(ii) $V(T_n) \rightarrow 0$, as $n \rightarrow \infty$.

Then T_n is a consistent estimator of $\gamma(\theta)$.

Q: Prove that in a sampling from $N(\mu, \sigma^2)$ population, the sample mean is a consistent estimator of μ .

Q: If $X_1, X_2, \dots, X_n$ are random observations on a Bernoulli variate X taking the value 1 with probability p and the value 0 with probability $(1 - p)$, show that: $\frac{\sum x_i}{n} \left(1 - \frac{\sum x_i}{n} \right)$ is a consistent estimator of $p(1 - p)$.

Practice questions

Q: A random sample X_1, X_2, X_3 of size 3 is drawn from a normal population with unknown mean μ and variance σ^2 . T_1, T_2, T_3 are the estimators used to estimate mean value $T_1 = X_1 + X_2 - X_3$,

$$T_2 = 2X_1 + 3X_3 - 4X_2, \quad T_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$$

- (A) Are T_1, T_2 are unbiased estimators.
- (B) For what value of λ , T_3 is unbiased?
- (C) With this value of λ , is T_3 a consistent estimator?

Efficient estimator

- If, of the two consistent estimators T_1, T_2 of a certain parameter θ , we have $V(T_1) < V(T_2)$, for all n then T_1 is more efficient than T_2 for all sample sizes.
- **Most efficient estimator:** If in a class of consistent estimators of a parameter, there exists one whose sampling variance is less than that of any such estimator, it is called the most efficient estimator.
- **Efficiency:** If T_1 is the most efficient estimator with variance V_1 and T_2 is any other estimator with variance V_2 then the efficiency η of T_2 is defined as : $\eta = \frac{V_1}{V_2}, 0 \leq \eta < 1$.

Q: A random sample X_1, X_2, X_3, X_4, X_5 of size 5 is drawn from a normal population with unknown mean μ . Consider the following estimators to estimate μ :

(i) $t_1 = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$ (ii) $t_2 = \frac{X_1 + X_2}{2} + X_3$ (iii) $t_3 = \frac{(2X_1 + X_2 + \lambda X_3)}{3}$, where λ is such that t_3 is an unbiased estimator of μ . Find λ . Are t_1 and t_2 unbiased? State giving reasons, the estimator which is best among t_1, t_2 and t_3 .

Q. Check whether $T_2 = \frac{2 \sum_{i=1}^n i x_i}{n(n+1)}$ is consistent estimator of population mean and also find, Of the estimators, $T_1 = \bar{X}$ and T_2 , which is more efficient estimator?

Practice question

Q: If X_1, X_2 and X_3 is a random sample of size 3 from a population with mean value μ and variance σ^2 . If t_1, t_2 and t_3 are the estimators used to estimate the mean value μ , where

$$t_1 = X_1 + X_2 - X_3, t_2 = 2X_1 + 3X_3 - 4X_2, \text{ and } t_3 = \frac{\lambda X_1 + X_2 + X_3}{3}.$$

- (i) Are t_1 and t_2 unbiased estimators?
- (ii) Find the value of λ such that t_3 is unbiased estimator for μ .
- (iii) With the value of λ , is t_3 a consistent estimator?
- (iv) Which is the best estimator?

Maximum Likelihood estimator (MLE)

Likelihood Function. *Definition.* Let $x_1, x_2, \dots, x_n$ be a random sample of size n from a population with density function $f(x, \theta)$. Then the likelihood function of the sample values $x_1, x_2, \dots, x_n$, usually denoted by $L = L(\theta)$ is their joint density function, given by

$$L = f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta) = \prod_{i=1}^n f(x_i, \theta). \quad (1)$$

The principle of maximum likelihood consists in finding an estimator for the unknown parameter $\theta = (\theta_1, \theta_2, \dots, \theta_k)$, say, which maximises the likelihood function $L(\theta)$ for variations in parameter *i.e.*, we wish to find $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k)$ so that

$$L(\hat{\theta}) > L(\theta) \quad \forall \theta \in \Theta$$

$$\text{i.e.,} \quad L(\hat{\theta}) = \text{Sup } L(\theta) \quad \forall \theta \in \Theta.$$

Thus if there exists a function $\hat{\theta} = \hat{\theta}(x_1, x_2, \dots, x_n)$ of the sample values which maximises L for variations in θ , then $\hat{\theta}$ is to be taken as an estimator of θ . $\hat{\theta}$ is usually called *Maximum Likelihood Estimator (M.L.E.)*. Thus $\hat{\theta}$ is the solution, if any, of

$$\frac{\partial L}{\partial \theta} = 0 \quad \text{and} \quad \frac{\partial^2 L}{\partial \theta^2} < 0 \quad \dots(15.54)$$

Since $L > 0$, and $\log L$ is a non-decreasing function of L ; L and $\log L$ attain their extreme values (maxima or minima) at the same value of $\hat{\theta}$. The first of the two equations in (15.54) can be rewritten as

$$\frac{1}{L} \cdot \frac{\partial L}{\partial \theta} = 0 \quad \Rightarrow \quad \frac{\partial \log L}{\partial \theta} = 0, \quad \dots(15.54a)$$

a form which is much more convenient from practical point of view.

Properties of MLE

Theorem 15.11. (Cramer-Rao Theorem). *“With probability approaching unity as $n \rightarrow \infty$, the likelihood equation $\frac{\partial}{\partial \theta} \log L = 0$, has a solution which converges in probability to the true value θ_0 ”. In other words M.L.E.'s are consistent.*

Remark. *MLE's are always consistent estimators but need not be unbiased. For example in sampling from $N(\mu, \sigma^2)$ population,*

MLE(μ) = $\bar{x}$ (sample mean), which is both unbiased and consistent estimator of μ .

MLE(σ^2) = s^2 (sample variance), which is consistent but not unbiased estimator of σ^2 .

Theorem 15.13. (Asymptotic Normality of MLE's). A consistent solution of the likelihood equation is asymptotically normally distributed about the true value θ_0 . Thus, $\hat{\theta}$ is asymptotically $N\left(\theta_0, \frac{1}{I(\theta_0)}\right)$ as $n \rightarrow \infty$.

Remark. Variance of M.L.E. is given by

$$V(\hat{\theta}) = \frac{1}{I(\theta)} = \frac{1}{\left[E \left(- \frac{\partial^2}{\partial \theta^2} \log L \right) \right]} \quad \dots(15.55)$$

Theorem 15.14. If M.L.E. exists, it is the most efficient in the class of such estimators.

Theorem 15.15. If a sufficient estimator exists, it is a function of the Maximum Likelihood Estimator.

Theorem 15.17. (Invariance Property of MLE). If T is the MLE of θ and $\psi(\theta)$ is one to one function of θ , then $\psi(T)$ is the MLE of $\psi(\theta)$.

Maximum likelihood estimator

Q: In random sampling from normal population $N(\mu; \sigma^2)$, find the maximum likelihood estimators for

(i) μ when σ^2 is known. (ii) σ^2 when μ is known. and (iii) the simultaneous estimation of μ and σ^2

Q: Find the maximum likelihood estimate for the parameter λ of a Poisson distribution on the basis of a sample of size n . Also find its variance.

It is known that a sample consisting of the values 12, 11.2, 13.5, 12.3, 13.8, and 11.9 comes from a population with the density function

$$f(x; \theta) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x > 1, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\theta > 0$. Find the maximum likelihood estimate of θ .

Practice question

Q. Prove that the maximum likelihood estimate of the parameter α of a population having density function: $\frac{2}{\alpha^2}(\alpha - x)$, $0 < x < \alpha$, for a sample of unit size is $2x$, x being the sample value. Show also that the estimate is biased.

🔍 Find the maximum likelihood estimator of parameter θ of a population having density function $(1 + \theta)x^\theta$.

Q: Suppose 10 rats are used in a biomedical study where they are injected with cancer cells and then given a cancer drug that is designed to increase their survival rate. The survival times, in months, are 14, 17, 27, 18, 12, 8, 22, 13, 19, and 12. Assume that the exponential distribution applies. Give a maximum likelihood estimate of the mean survival time.